COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	09/16/26	**DATE SUBMITTED:**	09/16/26
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ACTIVITY NO. 9.1 IMPLEMETING TREES

1. PROCEDURE OUTPUT

main.cpp > main()

```cpp
#include <iostream>
#include "binaryTree.h"


int main() {


    // Creating the root node
    BinaryTree<char>* root = new BinaryTree<char>('A');


    // Creating children of root node A
    root->insertLeftChild('B');
    root->insertRightChild('C');


    // Insert children to node B
    root->getLeftChild()->insertLeftChild('D');
    root->getLeftChild()->insertRightChild('F');


    // Insert child to node C
    root->getRightChild()->insertLeftChild('E');


    // Print the root node
    std::cout << "The Root: " << root->getRootVal() << std::endl;
    std::cout << "The Left child of root A: " << root->getLeftChild()->getRootVal() << std::endl;
    std::cout << "The Right child of root A: " << root->getRightChild()->getRootVal() << std::endl;


    std::cout << "The Left child of root B: "
         << root->getLeftChild()->getLeftChild()->getRootVal() << std::endl;


    std::cout << "The Right child of root B: "
         << root->getLeftChild()->getRightChild()->getRootVal() << std::endl;


    std::cout << "The Left child of root C: "
         << root->getRightChild()->getLeftChild()->getRootVal() << std::endl;


    return 0;

}

```

C binaryTree.h > ...

```h
#ifndef BINARYTREE_H
#define BINARYTREE_H


template <typename T>
class BinaryTree {
private:
    T key;


    BinaryTree<T>* leftChild;
    BinaryTree<T>* rightChild;


public:
    // Constructor
    BinaryTree(T rootObj);


    // Inserting child nodes
    void insertLeftChild(T newNode);
    void insertRightChild(T newNode);


    BinaryTree<T>* getLeftChild();
    BinaryTree<T>* getRightChild();


    T getRootVal();
};

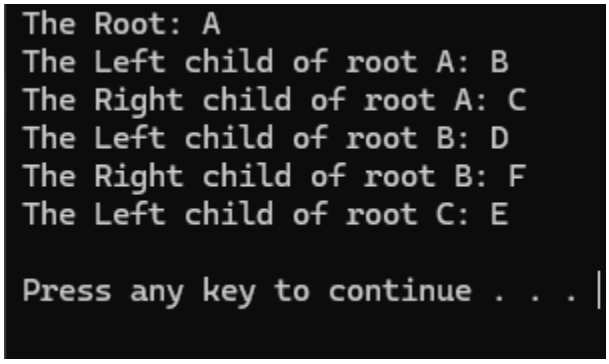
template <typename T>
BinaryTree<T>::BinaryTree(T rootObj) {
    key = rootObj;
    leftChild = nullptr;
    rightChild = nullptr;
}


template <typename T>
void BinaryTree<T>::insertLeftChild(T newNode) {
    leftChild = new BinaryTree<T>(newNode);
}

template <typename T>
void BinaryTree<T>::insertRightChild(T newNode) {
    rightChild = new BinaryTree<T>(newNode);
}
```

```
49
50  template <typename T>
51  BinaryTree<T>* BinaryTree<T>::getLeftChild() {
52      return leftChild;
53  }
54
55
56  template <typename T>
57  BinaryTree<T>* BinaryTree<T>::getRightChild() {
58      return rightChild;
59  }
60
61
62  template <typename T>
63  T BinaryTree<T>::getRootVal() {
64      return key;
65  }
66
67
68  #endif
69
```

Output:


The Root: A
The Left child of root A: B
The Right child of root A: C
The Left child of root B: D
The Right child of root B: F
The Left child of root C: E
Press any key to continue . . .

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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